# Week-6

**Name: G. Venkata Varun**

**Rollno: 2420030187**

**Adaptive Software Engineering – S6**

**Step 1: Check Docker Desktop**

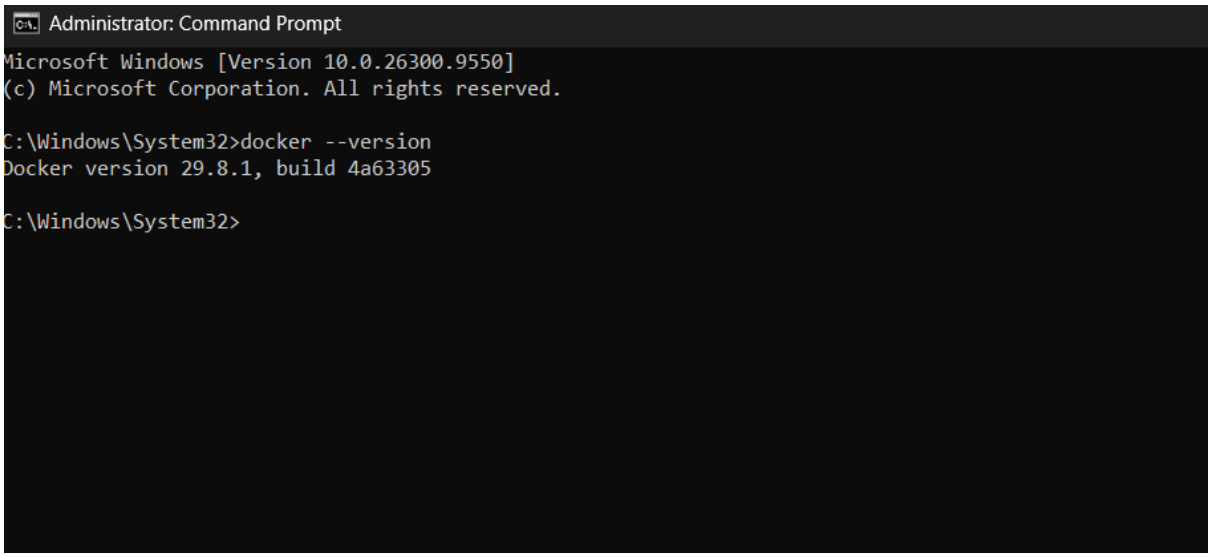
First, **open Docker Desktop** on your computer.

Wait until Docker Desktop shows that it is running.

Then open **PowerShell**.

Run:

docker --version




**Step 2: Create a project folder**

In PowerShell, go to a location where you want to create the project.

For example:

cd D:\

Create a folder:

mkdir docker-sample-app

Go inside it:

cd docker-sample-app

```
C:\Windows\System32>docker --version
Docker version 29.8.1, build 4a63305

C:\Windows\System32>cd C:\Users\vgami\Downloads\docker-sample-app
```

**Step 3: Create index.html**

Now type:

notepad index.html

Press **Enter**.

Notepad will open and may ask:

Do you want to create a new file?

Click **Yes**.

Paste this exactly:

<!DOCTYPE html>

<html>

<head>

  <title>Docker Sample Application</title>

</head>

<body>

  <h1>Hello from Docker!</h1>

  My first application is running inside a Docker container.

</body>

</html>

Then:

**File → Save**

Close Notepad.

**Step 4: Check that the file was created**

Back in CMD, type:

dir

You should see:

index.html

⚠️ **Important**

Make sure the file is named:

index.html

**not**

index.html.txt

```
C:\Windows\System32>cd C:\Users\vgami\Downloads\docker-sample-app

C:\Users\vgami\Downloads\docker-sample-app>notepad index.html

C:\Users\vgami\Downloads\docker-sample-app>dir
 Volume in drive C is Windows-SSD
 Volume Serial Number is 62C2-427B

 Directory of C:\Users\vgami\Downloads\docker-sample-app

12-10-2026  15:18    <DIR>          .
12-10-2026  15:17    <DIR>          ..
12-10-2026  15:18               218 index.html
               1 File(s)            218 bytes
               2 Dir(s)  290,337,533,952 bytes free

C:\Users\vgami\Downloads\docker-sample-app>
```

**Step 5: Create the Dockerfile**

Type:

notepad Dockerfile

Press **Enter**.

If Notepad asks whether you want to create a new file, click **Yes**.

**Step 6: Paste this Dockerfile**

Copy and paste exactly:

FROM nginx:latest

COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80

Then:

**File → Save**

Close Notepad.

**Step 7: Check the files**

Back in CMD, run:

dir

You should see **both**:

Dockerfile

index.html

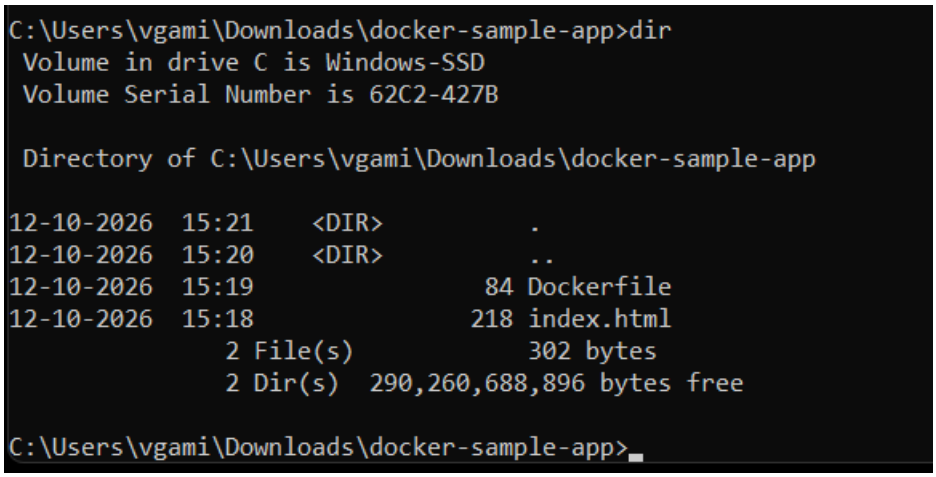
Your folder should now contain:

docker-sample-app

│

├── Dockerfile

└── index.html




**Step 8: Make sure you are in the project folder**

docker build -t docker-sample-app .

⚠️ **Important:** Keep the . at the end.

**Step 9: Wait for the build to finish**

You should see Docker doing things like:

[+] Building ...

[+] ...

At the end, you should see something similar to:

Successfully tagged docker-sample-app:latest

```
C:\Users\vgami\Downloads\docker-sample-app>docker build -t docker-sample-app .
[+] Building 67.3s (7/7) FINISHED                                                
```

xxxxxxxx docker-sample-app Up ... 0.0.0.0:8080->80/tcp

The important things are:

IMAGE: docker-sample-app

STATUS: Up

PORTS: 8080->80

```
C:\Users\vgami\Downloads\docker-sample-app>docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED          STATUS          PORTS                                       NAMES
a0281a60d91f   docker-sample-app  "/docker-entrypoint.…"   22 seconds ago   Up 21 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   docker-sample-container

C:\Users\vgami\Downloads\docker-sample-app>
```

**Step 13: Open your application**

Open **Chrome** and enter:

http://localhost:8080

You should see:

**Hello from Docker!**

and:

**My first application is running inside a Docker container.**




**Step 14: Check the container**

Go back to CMD and run:

docker ps

You should see your container:

docker-sample-container

with status:

Up ...

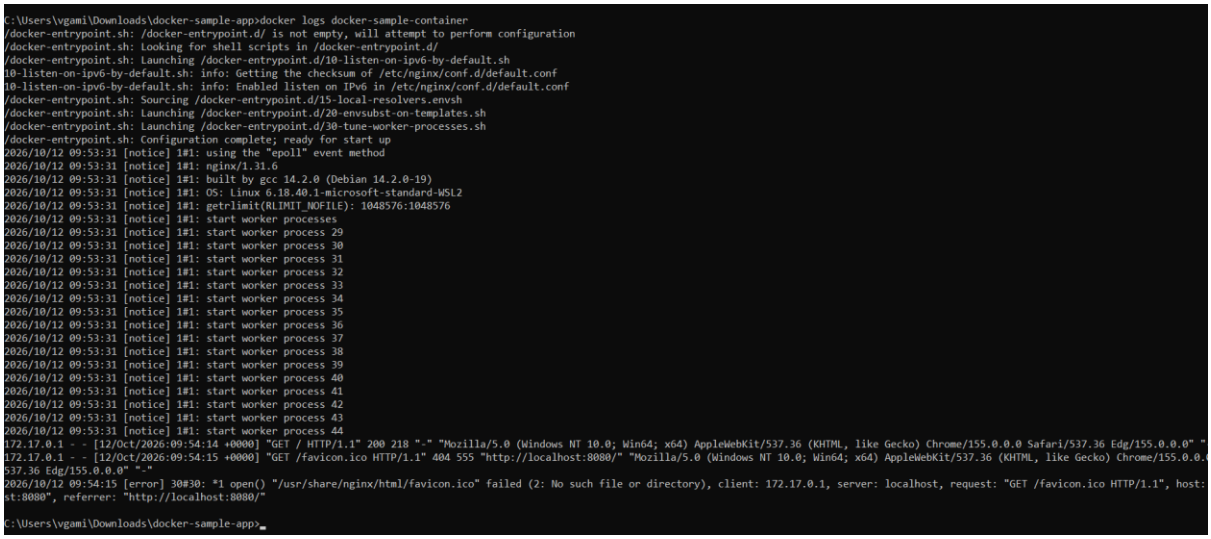
```
C:\Users\vgami\Downloads\docker-sample-app>docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED              STATUS              PORTS                                     NAMES
a0281a60d91f   docker-sample-app   "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   docker-sample-container
```

**Step 15: Check the container logs**

Run:

docker logs docker-sample-container

This displays the NGINX container logs.




**Step 16: Stop the container**

After you have verified the webpage, run:

docker stop docker-sample-container

You should get:

docker-sample-container

**Step 17: Check stopped containers**

Run:

docker ps -a

You should see:

docker-sample-container

with a status similar to:

Exited

```
C:\Users\vgami\Downloads\docker-sample-app>docker stop docker-sample-container
docker-sample-container

C:\Users\vgami\Downloads\docker-sample-app>docker ps -a
CONTAINER ID   IMAGE                                  COMMAND                  CREATED         STATUS                     PORTS     NAMES
a0281a60d91f   docker-sample-app                      "/docker-entrypoint.…"   2 minutes ago   Exited (0) 7 seconds ago             docker-sample-container
8944316e358d   sonarqube:latest                       "/opt/sonarqube/dock…"   7 days ago      Exited (130) 7 days ago              sonarqube
f4d8f5f51a81   gcr.io/k8s-minikube/kicbase:v0.0.51    "/usr/local/bin/entr…"   7 days ago      Exited (130) 7 days ago              minikube

C:\Users\vgami\Downloads\docker-sample-app>
```

